

Firestore Integration - Logic Map 4

Architecture Overview

The Firestore integration provides cloud-based backend services for the SMD Project.

Key Components:

1. Authentication Service

- Email/Password Authentication
- Google Sign-In Integration
- User Session Management
- Secure Token Handling

2. Realtime Database

- JSON-based data structure
- Real-time synchronization
- Offline persistence
- Security rules enforcement

3. Cloud Firestore

- Document and collection storage
- Advanced querying capabilities
- Transaction support
- Automatic indexing

4. Cloud Storage

- File upload and download
- Media handling
- Scalable storage
- Access control

Data Flow

User Request -> Kotlin Android App -> Firestore SDK

-> Authentication & Authorization -> Database Operations

-> Return Data to Client

Integration Points

- AndroidManifest.xml configuration
- google-services.json configuration file
- Firestore dependency initialization
- Security rules and access control